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Evaluating Large Language Models as Judicial Decision-Makers

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ABSTRACT

Large Language Models (LLMs) are increasingly shaping various domains, yet their ability to align with human judgment remains a critical challenge. This study explores the extent to which LLMs can serve as judicial decision-makers by comparing their sentencing decisions to those of 123 retired judges on two fictional cases involving rape and violence. We evaluate GPT, Gemini, and Claude using zero-shot, few-shot, and chain-of-thought prompts. LLMs showed greater consistency, producing significantly lower sentence disparity than judges. To assess accuracy, we treated the judges' average sentence as a conservative benchmark—acknowledging that the “correct” sentence is unknown. If models outperform even this minimal standard, they are closer to any plausible ground truth. Remarkably, all LLMs deviated less from the judges' mean than the judges themselves, suggesting that when properly prompted, LLMs can deliver more accurate sentencing decisions than human judges.

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Introduction

Marvin Frankel's landmark critique *Criminal Sentences: Law Without Order* (Frankel, 1973) sounded the alarm on the “lawless” inconsistency of criminal sentencing in the United States. His concerns were soon validated by empirical research, when studies showed that US judges impose very different sentences in similar cases (Austin and Williams, 1977; Clancy et al., 1981; Partridge & Eldridge, 1974).

Over the past several decades, policymakers have attempted to cure the harms of sentencing disparities through reforms such as structured sentencing guidelines. Frankel's critique of this lawlessness prompted Congress to adopt the Sentencing Reform Act of 1984, which created the United States Sentencing Commission and mandated guidelines intended to cabin judicial discretion and promote uniformity (Spohn, 2009). Various guideline regimes and mandatory sentencing laws proliferated across jurisdictions, explicitly aiming to reduce unwarranted variability in sentencing (Frase, 2005b). These measures have had some success in reducing sentencing disparities (Grunwald, 2015).

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However, structured guidelines and mandatory sentencing laws have come with notable drawbacks. Guidelines constrain judges' ability to consider the nuanced circumstances of particular defendants or cases, sometimes leading to mechanically calculated punishments that may be unjust in individual cases (Stith & Cabranes, 1998). They also increase prosecutorial power to influence sentences through charge and fact bargaining (Gazal-Ayal et al., 2013; O'Hear, 2005). They often lead to harsher sentences than the pre-guidelines regime (Hofer et al., 2004; Spohn, 2009). Moreover, while sentencing guidelines seem to have some effect on increasing uniformity, it is unclear to what extent the goal was achieved (Spohn, 2009).

Advancements in artificial intelligence—particularly the development of sophisticated large language models (LLMs)—present a promising opportunity to address sentencing disparities without resorting to rigid, inflexible guidelines. Unlike both structured sentencing guidelines, which constrain decisions through prescriptive ranges, and actuarial risk assessment instruments, which compress case facts into narrow numerical scores, LLMs can process nuanced narrative information and weigh a broader range of qualitative and contextual factors in ways that more closely resemble judicial reasoning. If such models can also enhance uniformity by ensuring that similar cases receive comparable sentences, while otherwise conforming with sentencing practices, they may help mitigate the “lawlessness” of sentencing.

In this study, we define “consistency” as the extent to which similar cases receive similar sentences. We focus on internal consistency—how much sentencing varies within a group of decision-makers (human judges or LLMs) when faced with the same case. We measure this by analyzing the variance of sentencing decisions.

Of course, consistency alone is not enough; sentencing must also reflect the unique circumstances of each case and adhere to legal principles. Sentences should not only be consistent but also accurate or “correct.” Yet, a core challenge in evaluating sentencing quality is the absence of an objective benchmark for what the correct sentence is. While existing sentencing practices—namely, the average sentences imposed by a large group of judges—are often used as a practical reference in structured sentencing regimes (Roberts et al., 2025), they do not necessarily represent a normative ideal. Still, if LLMs can generate decisions that are both more consistent and closer to this average than those of individual judges, they may reasonably be considered at least as accurate—if not more so—under conservative assumptions. Their value could be even greater in cases where human sentencing deviates from fair or proportionate outcomes.

To explore this possibility, we conducted an empirical study inspired by classic research on sentencing disparities, including the work of Partridge and Eldridge (1974), Austin and Williams III (1977), and Clancy et al. (1981). We analyzed sentencing decisions from 123 retired Israeli judges, each of whom was presented with detailed facts of two fictional cases and asked to determine the sentence they would impose under the given circumstances. The resulting dataset is publicly available for the research community.¹ Consistent with earlier findings, we found substantial variability in the

¹<https://osf.io/hv5s9>

sentences imposed by different judges for the same cases. We then applied various prompting techniques to elicit sentencing recommendations from multiple iterations of LLMs. Our analysis revealed that the AI-generated sentences closely matched the human average while exhibiting far lower variability, indicating greater internal consistency. These results suggest that LLMs may help reduce sentencing disparities without compromising accuracy by offering uniform recommendations that reflect existing judicial norms.

By examining this question, we hope to shed light on whether AI can enhance consistency in sentencing and how it might be integrated into broader efforts to promote more equitable sentencing outcomes, taking into account different practical and ethical considerations.

The remainder of this paper is structured as follows. Section 2 reviews the literature on sentencing disparities, structured sentencing reforms, and the integration of AI into judicial decision-making. Section 3 describes our methodological approach, including the cases, prompting strategies, and the process of collecting judicial decisions from human judges. Section 4 presents a comparative analysis of sentencing decisions made by human judges and LLMs. In Section 5, we discuss the broader implications of our findings for the use of AI in sentencing. Finally, Section 6 concludes the paper and outlines directions for future research.

Related Work

Sentencing Disparities: Extent and Sources

Sentencing disparity—where similarly situated defendants receive significantly different sentences—undermines fundamental principles of fairness, equality, and justice inherent in the criminal justice system (Frankel, 1973). Such variability can erode public confidence and trust in judicial processes, as it introduces an arbitrary element that conflicts with the ideal of consistent and equitable punishment. Thus, unwarranted sentencing disparity has been a long-standing concern of scholars and criminal justice reformers for many years (Kramer & Ulmer, 1996).

The problem of sentencing disparity has been documented extensively in legal scholarship and criminology for over half a century. Early quantitative studies focused on measuring overall inconsistency in sentences for similar cases, by presenting judges with identical case facts and requesting sentencing decisions. These vignette studies had the advantage of holding the facts constant for each judge, thus directly revealing inter-judge disparity that could not be attributed to case variation. In the Second Circuit's Sentencing Study, Partridge and Eldridge (1974) revealed substantial differences in the sentences rendered by different federal district court judges for the same offenses. The sentences ranged from 3 to 20 years of imprisonment for the same extortion and tax fraud offenses. When presented with the same possession of heroin case, judges imposed sentences ranging from 3 to 24 months. Austin and Williams III (1977) reported comparable results among state district court judges. For example, in a robbery case used in the study, the sentences ranged from 30 days to five years of imprisonment. Clancy et al. (1981) found similar disparities in a later study of federal judges. Studies using other methodologies also revealed the existence of sentencing disparities (Grunwald, 2015).

Structured Sentencing Reforms and Their Effects

In response to evidence of inconsistent sentencing, many jurisdictions adopted structured sentencing reforms aimed at constraining discretion and promoting uniformity. The paradigmatic and most rigid example is the U.S. Federal Sentencing Guidelines, implemented in 1987, which sought to create a more determinate sentencing system. Dozens of US states also introduced sentencing guideline systems or statutory sentencing ranges during the 1980s and 1990s (Frase, 2005b). These reforms were explicitly designed to reduce unwarranted disparity by “specifying the importance that each case and offender attribute should assume in arriving at a sentence” (Clancy et al., 1981)—essentially standardizing how judges weigh factors.

Studies generally confirm that guideline regimes have a positive but limited effect on sentence consistency. Most such studies leveraged the random assignment of cases to judges to examine whether sentences differ between judges and whether guidelines affect those differences. Anderson et al. (1999), Payne (1997), and Hofer et al. (1999) found that between-judges disparities were somewhat reduced after the introduction of the Federal Sentencing Guidelines. However, the latter two studies also found that disparities widened for certain types of offenses.

These modest achievements resulted from implementing the highly rigid federal sentencing guidelines. When the 2005 Booker decision (543 U.S. 220) made the federal guidelines advisory rather than mandatory, inter-judge disparities expanded once more. One study found that inter-judge sentencing disparities doubled following Booker (Yang, 2014).

Similarly, other guideline systems, including some within the United States, maintain greater flexibility by allowing judicial discretion, but this flexibility often comes at the cost of reduced consistency. A study of the effect of the Minnesota sentencing guidelines found that they initially reduced disparity in the decision to render a prison term but were less effective in determining whether prison was imposed in the first place, and this type of disparity increased over time (Frase, 2005a). A study in England and Wales also found a very limited reduction in sentencing disparities following the adoption of the Sentencing Guidelines there. Pina-Sánchez and Linacre (2016) report a 7.7% reduction in sentence length variance among matched assault cases, and even this limited increase in consistency did not correspond to the date of the guideline’s adoption. Hence, they were unable to prove a definitive causal link between the guideline and the observed improvements in consistency.

The evidence suggests that structured guidelines can somewhat reduce inter-judge disparity in sentencing outcomes, but they do not eliminate such disparities, and their effect diminishes over time. Moreover, inter-judge disparity is only one type of sentencing disparity. A judge may impose harsher sentences in some cases and more lenient ones in others, meaning that, on average, their sentences do not differ significantly from those of other judges, even if disparities persist within their own decisions (intra-judge disparity). Additionally, studies focusing on inter-judge disparity by comparing average sentences may overlook randomness in sentencing decisions. For instance, if one judge imposes a harsher sentence in a particular case and a more lenient one in another, while a second judge does the opposite, these inconsistencies may not be reflected in studies that only measure inter-judge averages. Since

before-and-after guideline studies have not systematically examined either intra-judge disparities or this form of case-specific randomness, it remains unclear whether and to what extent sentencing guidelines effectively reduce the full range of sentencing disparities.

More importantly, adopting a rigid guidelines regime comes with costs. Much has been written about the many limitations and unintended effects of guideline-based systems. A frequently voiced criticism is that strict guidelines overly constrain judicial discretion and can lead to “mechanical” justice. By design, guidelines curtail judges’ ability to individualize sentences based on unique case circumstances or compassionate judgment. This has prompted debates about fairness: while consistency is (partially) achieved, it may be at the expense of proportionality in individual cases (Frase, 2005b). Moreover, scholars point out that limiting judges’ discretion does not eliminate discretion from the system, but merely shifts it elsewhere. Prosecutors, who choose the charges and negotiate plea deals, exercise considerable influence over the applicable sentencing range under guidelines (Alschuler, 1978; Gazal-Ayal et al., 2013). If prosecutors use this power unevenly, disparities are simply relocated rather than eliminated—emerging earlier in the process through charge bargaining rather than at sentencing.

Another frequently observed consequence of structured sentencing regimes—particularly in the United States—has been an increase in sentence severity (Tiede, 2009). U.S. federal and many state guidelines have often steered courts toward longer custodial terms, prompting concerns that the pursuit of consistency may have produced disproportionately harsh outcomes and contributed to mass incarceration (Stith & Cabranes, 1998; Tonry, 1996). True, this pattern is not universal. Guideline systems in other jurisdictions, such as South Korea, Singapore, and England and Wales, have not consistently resulted in more severe sentencing outcomes (Roberts et al., 2025). Nevertheless, in the U.S. context, critiques of sentencing guidelines often emphasize their role in escalating severity.

All of the above led many to believe that the limited reductions in unwarranted disparity reached by the reliance on sentencing guidelines do not offset the associated loss (Waldfoegel, 1998).

Sentencing Law in Israel

In Israel, sentencing law does not prescribe numeric sentencing guidelines. Instead, for most offenses, the penal code establishes only maximum penalties and rarely specifies minimum sentences. Judges maintain significant discretion in determining sentences, guided primarily by statutory principles rather than structured numeric ranges. A significant reform adopted in 2012 introduced a structured approach to sentencing discretion, prioritizing proportionality or “just deserts” as the primary sentencing rationale. Under this approach, courts must first establish a proportionate sentencing range for the specific offense, considering the severity of the act, the harm it caused, and the offender’s culpability. Judges may deviate from this self-defined sentencing-range only for rehabilitation or incapacitation purposes. Other factors, unrelated to the offense, influence the sentence within the range. However, this

framework remains qualitative, as the legislation does not provide numeric baselines or presumptive sentence ranges to guide judicial discretion (Roberts & Gazal-Ayal, 2013).

Initially, the Israeli legislature considered introducing explicit “starting point sentences” to further structure judicial discretion, inspired by sentencing practices in jurisdictions such as England and Wales. These starting points were intended to offer judges a baseline from which deviations could be justified by specific aggravating or mitigating factors (Gazal-Ayal & Kannai, 2010). Ultimately, significant criticism and resistance from both legal practitioners and scholars, who raised concerns about excessive restrictions on judicial flexibility and the risk of mechanical sentencing, led to the abandonment of numeric guidelines in the final legislative enactment.

Given the absence of formal numeric guidelines, appellate review has remained the primary mechanism for reducing sentencing disparities in Israel, both before and after the reform. Prosecutors and defendants alike can appeal sentencing decisions, with appeals from district courts reaching the Supreme Court. Officially, appellate courts intervene only in cases of substantial judicial error or significant deviation from established sentencing policy. In practice, however, empirical studies show that the Supreme Court adopts a more interventionist approach, modifying the sentence in approximately one-quarter of the sentencing appeals it hears (Eisenberg et al., 2010). These appellate decisions establish binding precedents for lower courts and thus function as *de facto* sentencing guidelines aimed at promoting consistency. Nevertheless, despite this oversight, significant sentencing disparity persists in the Israeli criminal justice system, as we show below. In light of this, alternative methods for reducing sentencing disparities are needed. This study examines the application of LLMs as one such approach.

AI and Machine Learning in Judicial Decision-Making

Applications of AI and machine learning have rapidly expanded in judicial systems worldwide, particularly in criminal sentencing. The most common use cases involve decision support tools that assist judges with risk assessment and sentencing recommendations (Ryberg & Roberts, 2022). In the United States, for example, algorithmic risk assessment instruments are routinely used at sentencing to estimate a defendant’s likelihood of recidivism (Lippert-Rasmussen, 2022; U.S. Department of Justice, 2024). Widely deployed systems such as the COMPAS risk score and the Level of Service Inventory inform judges’ decisions on incarceration and supervision by providing data-driven risk evaluations (U.S. Department of Justice, 2024), although empirical evidence suggests that judges do not give such scores the expected weight in their decisions (Stevenson & Doleac, 2024). Beyond risk scores, a few jurisdictions have even piloted algorithmic sentencing recommendations for specific offenses. For example, some Malaysian courts introduced AI to recommend sentences for serious offenses such as rape and drug trafficking (Leong & Freeman, 2024). In China, judicial reforms have proclaimed ambitions to build “intelligent courts” that deeply integrate AI into case handling and sentencing. Some courts have already adopted AI tools to assist in sentencing (Shi, 2022). To date, fully automated sentencing by AI has not been adopted; the prevailing model is one of AI augmentation, where algorithms provide inputs or suggestions while human judges retain final authority (Ryberg, 2025).

The extent to which these computerized recommendations affect judges' decisions is disputed. For example, Virginia adopted an empirically derived regression-based instrument—the Nonviolent Risk Assessment (AVRA) tool—which recommends alternative sanctions instead of incarceration (Garrett et al., 2018). Ho et al. (2023) show that adopting this tool increases the likelihood of offering alternative punishments, lowers the probability of incarceration, and reduces sentence terms. Stevenson and Doleac (2024) also found that while the NVRA altered judges' sentencing practices, judges' discretion mitigated this change and did not improve public safety or reduce incarceration rates.

The extent to which these computerized sentencing tools improve sentencing is highly debated. Proponents of these sentencing assistance tools argue that such technologies can significantly enhance judicial decision-making by increasing consistency, transparency, and efficiency in sentencing practices. AI-driven risk assessment algorithms aim to standardize judicial outcomes by providing data-informed insights into defendants' recidivism risks, potentially reducing unwarranted disparities among judges (Roberts & Schwarze, 2021). Moreover, judges have been shown to be consistently inaccurate in predicting which offenders are likely to reoffend, and AI instruments have shown greater accuracy, speed, and consistency (Bagaric & Hunter, 2022). And while these tools may exhibit biases against minorities, such biases may be more readily identified and corrected in AI systems than in human judgment (Bagaric et al., 2022). For instance, AI systems can be systematically reprogrammed to align with evolving policy standards by instructing models to exclude specific factors such as race or socioeconomic status, whereas changing entrenched decision-making patterns among human judges is notoriously difficult (Mullainathan, 2019).

In contrast, critics of AI sentencing tools emphasize significant ethical, practical, and fairness concerns. One major concern involves automation bias, where judges might uncritically defer to algorithmic recommendations, thereby eroding judicial discretion and potentially undermining the individualized and context-sensitive nature of justice (Zerilli, 2021). Another concern is that reliance on AI risks politically motivated misuse, possibly reinforcing punitive policies through a veneer of algorithmic neutrality, thus calling for robust oversight mechanisms and governance frameworks to safeguard justice (Thomsen, 2022). Furthermore, it is argued that AI systems cannot reliably account for the full complexity of case-specific sentencing factors, and this input problem undermines the quality of the output (Ryberg, 2025).

The debate does not end here. Scholars debate whether the use of such tools mitigates or exacerbates biases against racial and other economically disadvantaged groups (Bagaric et al., 2022; Chouldechova, 2017; Davies & Douglas, 2022). The issue of transparency is also debated, with some believing that the opacity of AI tools undermines their legitimacy, and others arguing that these tools can improve transparency compared to human judges (Chiao, 2022; Ryberg, 2022; Ryberg & Petersen, 2022). Weighing the advantages and disadvantages of AI in criminal sentencing, Schwarze and Roberts (2022) argue that AI should supplement rather than supplant judicial decision-making, serving as an advisory tool that enhances consistency and reduces bias while preserving the essential human elements of the sentencing process.

Most of the literature examined thus far has focused on traditional algorithmic approaches to sentencing, such as risk assessment tools and statistical models that

operate on structured data with predetermined variables. These traditional tools may indeed offer consistency, in the sense that they produce the same result when the structured input is identical. However, they rely heavily on coded variables, which can fail to capture the full nuance and complexity inherent in sentencing decisions. Important contextual details—such as the nature of a defendant’s remorse, the subtleties of the offense circumstances, or mitigating social factors—often reside in unstructured textual materials rather than in structured fields. Manual coding of such narratives into discrete variables inevitably introduces subjectivity and results in information loss. Thus, while traditional algorithmic tools may achieve internal consistency, they often lack the flexibility and sensitivity required for nuanced legal decision-making.

LLMs represent a fundamentally different paradigm of artificial intelligence with distinct capabilities and limitations that warrant separate consideration in the sentencing context.

Large Language Models in Judicial Sentencing

LLMs are a recent entrant into judicial decision-making, and courts are cautiously beginning to explore their potential. One distinguishing feature of LLMs is their ability to process unstructured textual input, such as the materials judges typically review (e.g. indictments, probation reports, and parties’ arguments). Their ability to generate coherent and contextually appropriate text further enhances their appeal. While courts are beginning to explore the utility of these models, particularly for tasks involving text processing and generation (Lady Chief Justice et al., 2023), their role in substantive legal reasoning and decision-making, such as sentencing, remains a subject of cautious consideration (de la Osa & Remolina, 2024).

Emerging research explores the integration of LLMs into various aspects of the judicial process, with a focus on understanding how these models might assist judges in their decision-making. One notable interaction pattern observed in Shenzhen, China, involves judges making initial decisions, followed by an LLM generating reasoning to support these decisions, which the judges then revise to produce the final judgment (Liu & Li, 2024). This suggests a potential role for LLMs in enhancing the articulation and transparency of judicial reasoning rather than directly determining outcomes. A vignette study with Chinese judges failed to find a significant effect of using LLMs on the influence of an extra-legal variable in verdicts and sentencing (Liu & Li, 2024).

Posner and Saran (2025) directly compared the decisions of GPT-4o to those of U.S. federal judges using a controlled experimental design. They found that the LLM aligned more closely with student participants than with judges, demonstrating greater adherence to precedent and less responsiveness to extralegal factors such as the defendant’s sympathetic portrayal.

Other studies examined the accuracy of LLMs in predicting judicial decisions. For example, two studies developed such models and examined their ability to predict decisions of the European Court of Human Rights (ECHR) (Samee et al., 2024; Wang et al., 2024). In addition, Farber (2024) found that while LLMs underestimated sentence lengths in Israeli terrorism cases, they showed strong internal consistency and

predictive alignment with actual sentences, highlighting their potential as decision-support tools.

To the best of our knowledge, no study to date has analyzed the decisions of LLMs and many human judges on a similar sentencing case vignette. This approach enables a direct comparison of the consistency and accuracy of decisions across a large group of human judges and multiple LLMs.

Method

We present an overview of our methodology in [Figure 1](#). First, in Section 3.1, we describe the fictional cases used to evaluate LLMs and human judges. Next, in Section 3.2, we outline our approach to evaluating these cases using various LLMs and prompt engineering strategies. Section 3.3 presents the human judgment process. Finally, we compare and analyze the results obtained from human judges and LLMs.

Data Collection

We employ two fictional cases for evaluation. These cases were originally developed in the original language and utilized by Gazal-Ayal et al. (2025) to analyze sentencing disparity among retired judges. The translated versions of the two cases are provided in full in [Appendix A](#). Using fictional cases ensures that the LLMs have not been exposed to the data beforehand.

Both cases involved charge and fact bargaining. By using plea agreements, we ensured that the judges had full access to all relevant facts, mirroring the information typically available to judges in real cases following a plea deal. Thus, judges could not attribute variations in sentencing decisions to incomplete evidence or insufficient details commonly seen during court hearings. The judges were provided with all agreed-upon facts, the relevant content of the probation report, and the specific sentencing requests from both the prosecution and the defense. Each party referenced two relevant court precedents for each case to support these requests.

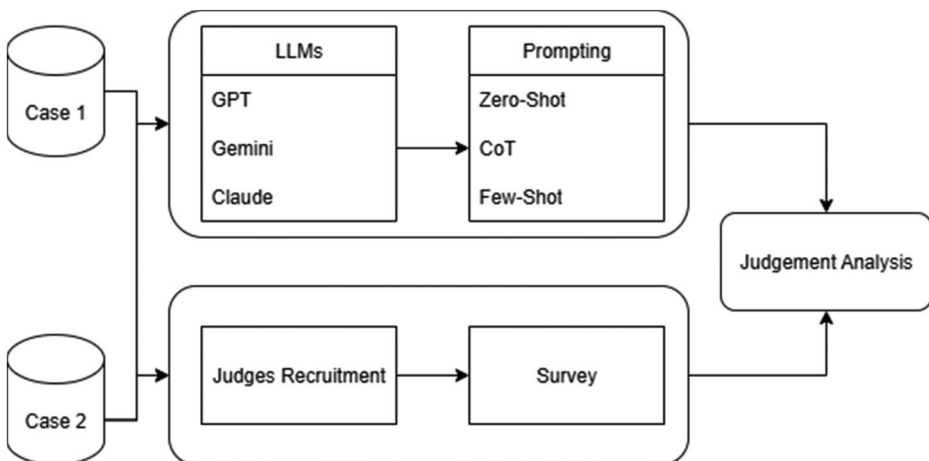


Figure 1. Overview of our method for comparing LLMs with human judges.

Case 1 involved a violent offense in which the defendant committed aggravated assault against another man during an alcohol-induced altercation, resulting in severe injuries. The prosecution requested a four-year prison sentence, emphasizing the gravity of the harm caused. The defense, in contrast, sought a one-year sentence, citing the defendant's personal mitigating circumstances and prospects for rehabilitation. Both sides supported their positions with relevant examples of sentences issued in other (real) cases, as is commonly done in sentencing hearings.

Case 2 involved a sexual offense in which the defendant assaulted a female acquaintance after a consensual interaction turned non-consensual. The prosecution sought a five-year prison sentence, citing court precedents that underscored the gravity and societal impact of sexual offenses. The defense, by contrast, advocated for a 15-month sentence, relying on Supreme Court sentencing decisions that highlighted the defendant's remorse, lack of prior convictions, and potential for rehabilitation.

We deliberately selected cases where a non-prison sentence was highly unlikely. This design allowed us to concentrate on variation in prison sentence length without conflating it with binary in/out decisions. Combining these two types of outcomes would complicate statistical comparison between judges and LLMs, reducing clarity and interpretability. By focusing on custodial terms, we provide a cleaner test of consistency in sentencing.

LLMs and Prompt Strategies

To evaluate LLM-based sentencing predictions, we used multiple models and applied various prompt engineering techniques. We tested the following models: ChatGPT (GPT-4o), Gemini (Gemini-2.0-Flash), and Claude (Claude-3.5-Sonnet). All outputs were generated through the providers' official APIs (OpenAI, Google, Anthropic), which enabled more precise parameter control, ensured complete independence between runs, and minimized the potential for human or contextual bias. Each API call was stateless, meaning no conversational history was retained between runs. This setup allowed automated execution of multiple independent iterations per case, improved procedural consistency, and eliminated possible errors associated with manual prompting (e.g. formatting or resubmission errors). API-based evaluation is widely recognized as a best practice for ensuring scalability and reproducibility in LLM research across diverse domains (Bang et al., 2023; Chakraborty et al., 2025; Rawashdeh et al., 2024), including legal analysis (Dahl et al., 2024).

For each LLM we employ the same system prompt: You are a judge in a criminal proceeding in Israel.

Each model was assessed using the following prompt strategies:

- **Zero-shot prompting**—Each LLM was presented with the case and asked to determine an appropriate sentence without prior examples. The zero-shot prompt used in our experiments was: Attached are two cases describing sentencing arguments in different legal proceedings. Please specify the sentence you would impose on the defendants in these cases. (We acknowledge that the information is presented in summary form—please do your best to

determine an appropriate sentence based on the available details. There is, of course, no single correct answer.)

- **Few-shot prompting**—Each LLM was provided with three similar cases, each with a corresponding sentence, to guide its response. The examples were selected by a domain expert—an experienced researcher from the Faculty of Law. The domain expert identified ten real cases similar to each of our research cases. To mitigate bias and adhere to token limitations, we selected three cases based on sentence-length distribution: For Case 1 (violent offense): 9 months, 28 months, and 4 years; for Case 2 (sexual offense): 9 months, 5 years, and 8 years. This selection ensures that the LLMs are not skewed toward overly lenient or harsh examples. The full text of all examples used in this strategy is available in [Appendix B](#).
- **Chain-of-thought (CoT) prompting**—Each LLM was prompted to provide a step-by-step reasoning process before determining the sentence. The CoT prompt combined the zero-shot prompt with an explicit instruction to reason step by step: Solve the following problem step by step. Explain each stage of your reasoning before reaching a final conclusion.

Each model was executed 30 times per prompt strategy with the temperature parameter set to zero and to the default. For Case 2, when the LLM responses did not include the sentencing decision (due to token limitation), we modified the prompt to have the model start its response with Case 2 instead of Case 1 and ran an additional 30 iterations. When a model failed to provide an explicit sentence in exact years of imprisonment (e.g. only offering a range), additional iterations were performed until a valid sentencing decision was obtained.

Human Judges

We analyzed sentencing decisions made by retired judges who were provided with the same detailed descriptions of the cases and sentencing arguments through questionnaires. The judges were tasked with determining an appropriate sentence for each case based on the information provided, which included, as stated above, the facts of the case, details of the probation report, and the sentencing arguments of both parties.

The study received ethical approval from the Institutional Review Board.² A total of 123 retired judges participated, comprising 87 men (70.7%) and 36 women (29.3%) from various judicial instances. Their experience in sentencing varied: 90 judges had extensive sentencing experience, 25 had limited experience, and 7 had no prior sentencing experience.

Their professional backgrounds included 36 former prosecutors, 34 former defense attorneys, and 61 former civil law attorneys. Regarding their final judicial roles before retirement: 53 served in magistrate courts, 57 in district courts, 5 in the Supreme Court, and 7 in other courts.

²Approval number 002/2020, University of Haifa, Faculty of Law, Review Board.

The analysis compared their sentencing tendencies with those generated by LLMs. To determine whether LLMs are superior sentencers, we tested two accumulative conditions: first, whether LLMs are more consistent than judges, and second, whether their sentence recommendations are, on average, more accurate.

Results

In both cases, all human judges and all LLMs imposed prison sentences.

Case 1: Sentencing Outcomes for Violent Offense

We assess the alignment of LLM-generated sentencing outcomes with human judicial decisions by analyzing the distribution of sentencing predictions for Case 1. Figures 2 and 3 provide visual representations of these outcomes through a boxplot and a distribution plot, respectively.

We observe that all LLMs, except for GPT in the zero-shot setting and Gemini in the few-shot setting, produce consistent sentencing outcomes. In contrast, human judges exhibit a broader range of sentencing years, with substantial variability in their decisions. These findings suggest that LLMs tend to generate sentencing outputs with less variability. Interestingly, most LLMs frequently recommend sentences of 2 or 2.5 years in prison, indicating a strong preference for certain sentencing outcomes.

Focusing on the overall sentencing distributions (Figure 3) reveals additional insights. Human judges display a more dispersed sentencing pattern, with a higher standard deviation than any of the LLMs. This variation could be attributed to inter-judge disparity. In contrast, LLM-generated sentences tend to cluster around specific values, suggesting a narrower decision space.

To further examine the results, we compare the average sentence lengths and standard deviations between LLMs and human judges, as shown in Table 1. The human judges impose an average sentence of 2.14 years, with a standard deviation

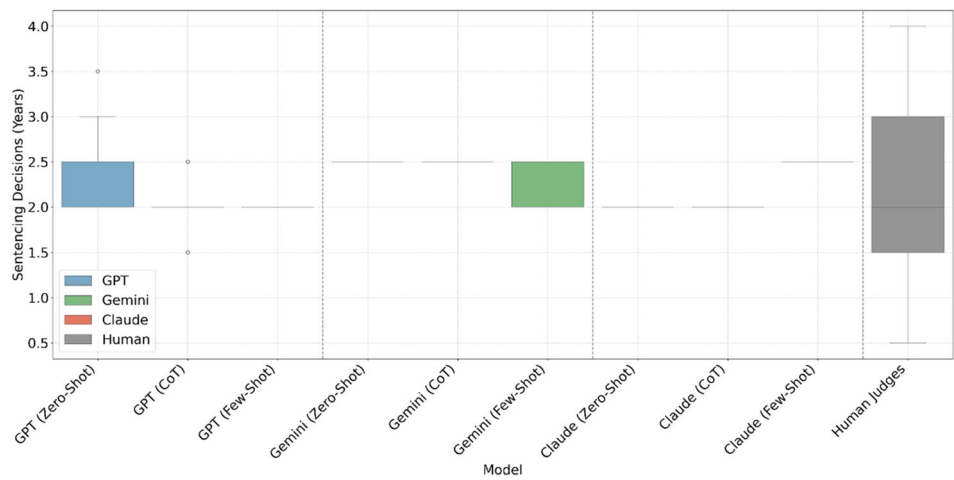


Figure 2. Boxplot representation of sentencing decisions for Case 1.

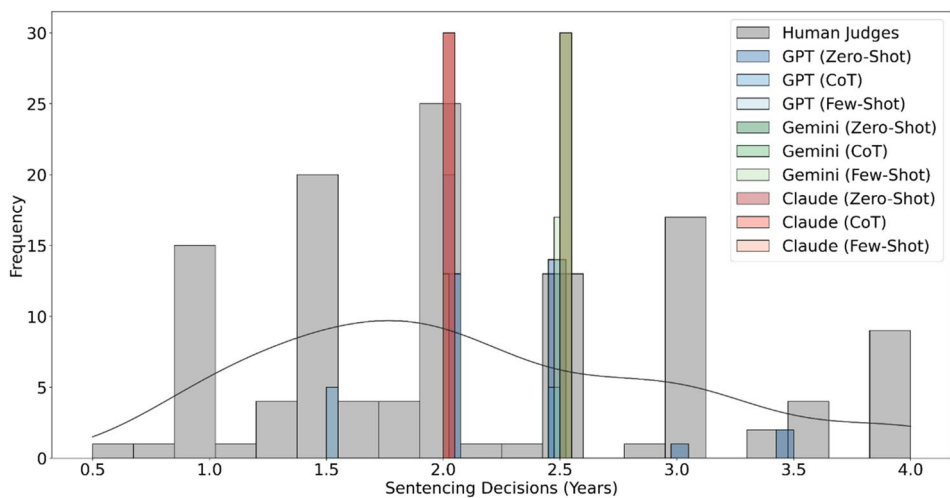


Figure 3. Distribution of sentencing decisions for Case 1.

of 0.87, reflecting variability in judicial discretion. Among the LLMs, Claude demonstrates the most stable predictions, with a standard deviation of zero for all examined prompting strategies. Notably, without any few-shot examples, Claude consistently assigns a 2-year sentence. However, when provided with examples as part of the few-shot strategy, Claude shifts toward a harsher punishment, increasing the sentence to 2.5 years. In contrast, GPT and Gemini exhibit an opposite trend—assigning relatively severe or equivalent sentences in the zero-shot setting, but reducing their average sentence length when exposed to few-shot examples. This divergence suggests that few-shot learning does not impose a uniform directional shift and interacts differently with each LLM’s internal decision-making process.

To evaluate the statistical significance of differences between LLM-generated and human sentencing decisions, we apply the Mann-Whitney U test. This rank-based non-parametric test determines whether two independent groups differ significantly in their distributions. Unlike parametric tests, the Mann-Whitney U test does not assume normality in the data, making it well-suited for comparing human and LLM-generated sentencing decisions, which may follow different distributional patterns. A low U-statistic combined with a p -value below 0.05 indicates that the LLM’s sentencing predictions significantly differ from human judges. In contrast, a high U statistic and a p -value above 0.05 suggest no statistically significant difference, implying that the LLM’s sentencing pattern approximates that of human judges.

In addition to the Mann-Whitney U test, we employed Levene’s test to assess the homogeneity of variances between LLM-generated and human sentencing outcomes. Levene’s test evaluates whether the variances of two or more groups are statistically different, providing insight into the consistency or variability of sentencing decisions. A significant result ($p < 0.05$) indicates that the variance of one group differs significantly from the other, pointing to differing levels of dispersion. As shown in Table 1, all LLMs exhibit significantly lower variance compared to human judges ($p < 0.05$ in most cases). These results reinforce the conclusion that LLMs generate more consistent, less variable sentences than human judges.

Table 1. Case 1—Comparison of sentencing statistics between LLMs and human judges, including mean sentence length, standard deviation, Mann-Whitney U test results, and Levene’s test results.

LLM	Prompting strategy	Average	Standard deviation	U statistic	p-Value	Levene statistic	p-Value
GPT	Zero-Shot	2.3666	0.4138	2305	0.0324	18.9	<0.01
	CoT	2	0.2936	1827.5	0.9366	36	<0.01
	Few-Shot	2	0	1875	0.8892	64.2	<0.01
Gemini	Zero-Shot	2.5	0	2505	0.0020	64.2	<0.01
	CoT	2.5	0	2505	0.0020	64.2	<0.01
	Few-Shot	2.2833	0.2520	2232	0.0716	28	<0.01
Claude	Zero-Shot	2	0	1875	0.8892	64.2	<0.01
	CoT	2	0	1875	0.8892	64.2	<0.01
	Few-Shot	2.5	0	2505	0.0020	64.2	<0.01
Human Judges		2.1456	0.8782	–	–	–	–

Examining [Table 1](#), we found that Gemini (zero-shot and CoT) and Claude (few-shot) exhibit statistically significant deviations from human judges, as indicated by their low p -values. This suggests that these models generate sentencing patterns that substantially differ from judicial decisions. On the other hand, GPT in the CoT and few-shot settings, as well as Claude in the zero-shot and CoT settings, produce sentencing distributions that are statistically indistinguishable from human judges. These findings indicate that the choice of LLM and prompting strategy plays a crucial role in determining sentencing alignment with human judicial decisions.

Case 2: Sentencing Outcomes for Sexual Offense

We illustrate the sentencing outcomes for Case 2, depicting sentencing variations across LLMs and human judges in [Figures 4](#) and [5](#). The statistical results are detailed in [Table 2](#), including the average sentence, standard deviation, and Mann-Whitney U test results.

Examining the sentencing distributions for Case 2, we observe that LLM-generated sentences tend to be more concentrated around specific values, whereas human judges demonstrate greater variability in their sentencing decisions. The human judges impose an average sentence of 2.78 years, with a standard deviation of 1.14, reflecting a wide range of discretionary decisions. Most LLMs produce more rigid sentencing patterns, often converging around 3 years, with limited variation.

Notably, most LLM configurations produce similar sentencing outcomes to those of human judges. Specifically, GPT and Gemini under all prompting strategies yield sentence distributions that do not significantly differ from human judicial decisions. In contrast, only Claude in the few-shot setting shows a clear deviation from human judges, with a higher average sentence length and a statistically significant difference in the Mann–Whitney U test ($p < 0.01$). Taken together, these findings indicate that, in this case, most LLMs generate judgments that are aligned with human sentencing decisions.

To further evaluate variability across models, we conducted Levene’s test to assess whether the variance in sentencing predictions differs significantly between LLMs and human judges. Levene’s test results, shown in [Table 2](#), reveal that most LLMs exhibit significantly lower variance than human judges ($p < 0.01$). However, one notable

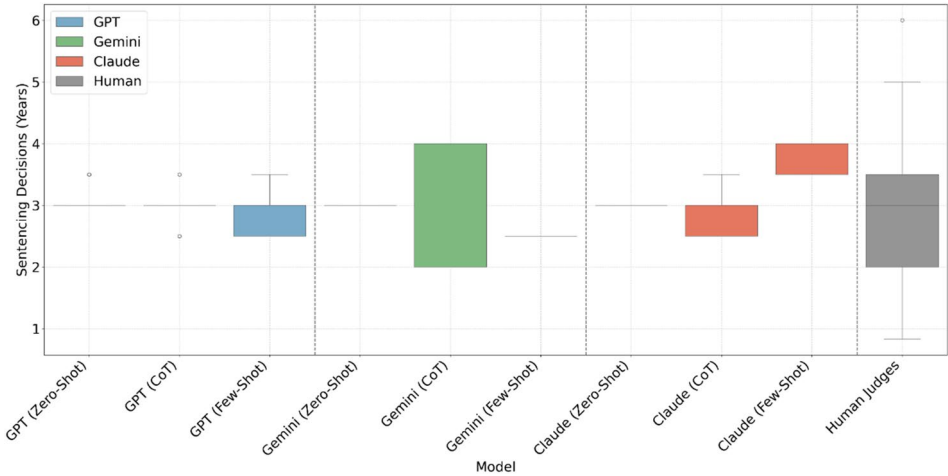


Figure 4. Boxplot representation of sentencing decisions for Case 2.

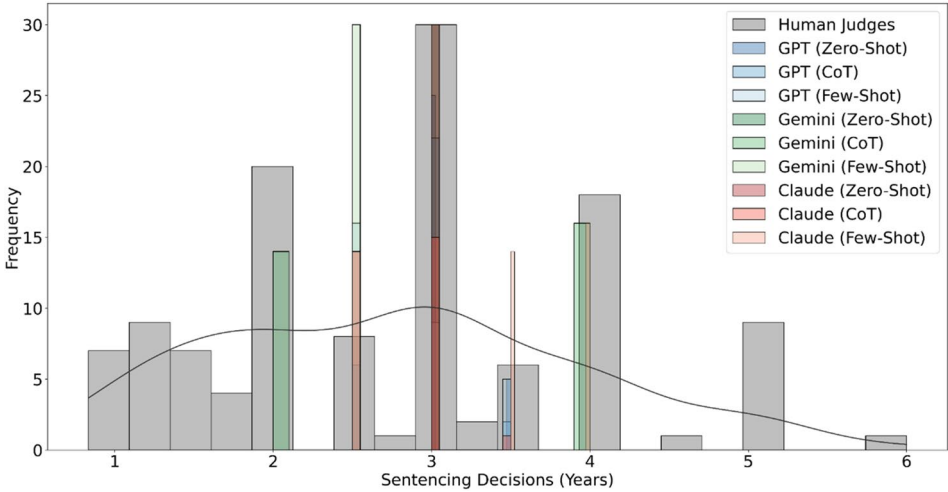


Figure 5. Distribution of sentencing decisions for Case 2.

Table 2. Case 2—Comparison of sentencing statistics between LLMs and human judges, including mean sentence length, standard deviation, Mann-Whitney U test results, and Levene's test results.

LLM	Prompting strategy	Average	Standard deviation	U statistic	p-Value	Levene statistic	p-Value
GPT	Zero-Shot	3.0833	0.1895	2242.5	0.0614	44.2	<0.01
	CoT	2.9333	0.2537	2061	0.3115	40.12	<0.01
	Few-Shot	2.8166	0.3824	1914.5	0.7482	24.8	<0.01
Gemini	Zero-Shot	3	0	2145	0.1553	61.3	<0.01
	CoT	3.0666	1.0148	2166	0.1349	0.19	0.6597
	Few-Shot	2.5	0	1530	0.1429	61.3	<0.01
Claude	Zero-Shot	3	0	2145	0.1553	61.3	<0.01
	CoT	2.7833	0.2841	1877.5	0.8812	31.6	<0.01
	Few-Shot	3.7666	0.2537	2922	<0.01	33.2	<0.01
Human Judges		2.7879	1.1475	—	—	—	—

exception is Gemini in the CoT setting, which exhibits a higher standard deviation (1.01), and a Levene’s test that does not indicate a statistically significant difference ($p=0.65$). This result increase in dispersion suggests that while most LLMs produce tightly clustered sentences, Gemini with CoT may incorporate more diverse reasoning paths, resulting in a broader spread of outcomes.

Accuracy—Alignment with Human Judges

While the preceding analysis demonstrates that LLMs exhibit greater consistency than human judges, consistency alone does not establish superior performance. A hypothetical system that mechanically assigns the same sentence to every case, regardless of its specific facts, would be perfectly consistent yet entirely useless. Similarly, selecting model parameters solely to minimize output variability, without regard to the substantive quality of those outputs, would produce artificially inflated consistency metrics that bear no relation to actual decision-making competence. Therefore, the critical question is not simply whether LLMs are consistent, but whether their consistent outputs cluster around appropriate sentencing values. In other words, we must evaluate both consistency and accuracy together. The analysis below examines whether LLM-generated sentences not only exhibit lower variability than human judgments but also align more closely with a reasonable benchmark for sentencing appropriateness.

To evaluate accuracy, we must establish a benchmark for comparison. Since there is no objectively known or observable “correct” sentence in these cases, we use the average sentence imposed by the human judges as our reference point. This choice represents a conservative test that is maximally favorable to the judges themselves: if LLMs deviate less from the judges’ mean than judges deviate from their own mean, this demonstrates superior accuracy regardless of what the true correct sentence might be. In other words, if LLMs cluster more tightly around the judicial average than judges do around their own average, they exhibit less decision-making variance from any reasonable reference point. For each LLM-generated sentence, we calculated its mean absolute deviation (MAD) from the human average. We then assessed the variability of these deviations by comparing their standard deviations to those of the human judges using Levene’s test. The results are presented in [Table 3](#).

Table 3. Levene’s test results comparing the variance of each LLM’s deviation from the human mean to the variance among human judges.

LLM	Prompting strategy	Case 1			Case 2		
		MAD	Levene statistic	<i>p</i> -Value	MAD	Levene statistic	<i>p</i> -Value
GPT	Zero-Shot	0.347	15.3	<0.01	0.295	37.7	<0.01
	CoT	0.263	20.4	<0.01	0.260	51.7	<0.01
	Few-Shot	0.145	57.5	<0.01	0.335	40.4	<0.01
Gemini	Zero-Shot	0.354	57.5	<0.01	0.212	69.7	<0.01
	CoT	0.354	57.5	<0.01	1.014	26.3	<0.01
	Few-Shot	0.263	31.7	<0.01	0.287	69.7	<0.01
Claude	Zero-Shot	0.145	57.5	<0.01	0.212	69.7	<0.01
	CoT	0.145	57.5	<0.01	0.264	56.4	<0.01
	Few-Shot	0.354	57.5	<0.01	0.978	20.7	<0.01
Human Judges		0.723	–	–	0.942	–	–

As shown in the table, in both cases, all LLM variants exhibit significantly lower variance in their deviations from the human mean compared to the human judges, as evidenced by consistently high Levene statistics and significant p -values ($p < 0.01$). This holds across all prompting strategies and models for both Case 1 and Case 2. Notably, in Case 2, Gemini under CoT prompting and Claude under few-shot prompting exhibit high MAD values; however, their deviation variances still remain significantly lower than those of human judges ($p < 0.01$).

To assess the collective consistency and accuracy of LLMs relative to human judges, we aggregated the outputs from all evaluated models (GPT, Claude, Gemini) and prompting strategies (zero-shot, chain-of-thought, and few-shot) into a single distribution. We then calculated the absolute deviation of each LLM-generated sentence from the average sentence imposed by the human judges, treating that average as a conservative proxy for the correct sentencing decision. A Levene's test revealed a statistically significant difference in the variance of these deviations between LLM outputs and human decisions: for the first case, Levene's statistic = 205.1, $p < 0.01$; for the second case, Levene's statistic = 93.06, $p < 0.01$.

These findings show that, also when pooling outputs across models and prompting strategies, LLMs generated sentencing decisions that were both more consistent and more accurate than those of individual judges.

As noted above, our choice to use the judges' average as the benchmark is conservative. If the true correct sentence differs from this average, the advantage of the LLMs in accuracy would be even greater, since treating the judges' average as the correct sentence minimizes the judges' deviation by definition.

Robustness and Sensitivity Analysis

Effect of Temperature Settings- As noted by Posner and Saran (2025), temperature plays a key role in configuring LLMs and should be adjusted based on the use case. Thus, we evaluated the influence of temperature on model consistency using the mean absolute deviation (MAD) and Levene's statistics, with the results for the default temperature settings presented in Table 4. Consistent with our main findings, most LLMs achieved substantially lower MAD values than human judges in both cases, indicating accurate sentencing decisions. Among the evaluated models, Claude exhibits the highest accuracy in both cases, achieving its strongest performance under the CoT prompting strategy in Case 1 and under zero-shot prompting in Case 2. These results indicate that even when the temperature was increased to the models' default settings, sentencing decisions remained accurate, yielding outcomes statistically comparable to those obtained at zero temperature. Overall, while higher temperatures introduce greater stochasticity, the models' reliability remains robust and, in many cases, exceeds that of human judges, as reflected by their lower MAD values. We also present the average sentence length and standard deviation for each LLM under zero temperature, along with the default temperature, together with the Mann-Whitney results in Appendix C.

Effects of Sentence Increments—We further examined whether the observed consistency of LLM outputs was merely an artifact of their tendency to recommend sentences in six-month increments. Our data reveal that human judges follow a similar convention: only 15.4% of judgments in Case 1 and 14.6% in Case 2 were not

expressed in half-year increments. Crucially, rounding all human sentences to the nearest six-month interval resulted in virtually no change to their standard deviation. This indicates that the models’ stability is not a product of numerical rounding, but rather reflects a fundamentally more consistent weighting of case facts.

Sensitivity to factual changes—we also tested whether LLMs produce consistent sentences when the facts of the case change. To do this, we compared the outputs of each LLM and prompting strategy for the original Case 1 vignette and a modified version of the case in which the facts were altered.

In the modified version, one main key change was made; the nature of the violent act was significantly softened: in the original case, the defendant actually stabbed the victim with a broken bottle, causing severe injuries and hospitalization, whereas in the modified case, he only threatened to stab, without inflicting physical injuries. Collectively, this edit reduced both the severity of the offense and its consequences, thus lowering the overall gravity of the case.

Significantly, we did not alter the sentence recommendations presented by the prosecution or the defence. This ensured that any changes in the models’ outputs reflected sensitivity to the factual circumstances of the case rather than simple alignment with the numeric requests.

Table 5 reports the averages, standard deviations, and Mann–Whitney results for this comparison. In every case, the *p*-values are extremely small, indicating decisive

Table 4. Levene’s test results comparing the variance of each LLM’s (with default temperature) deviation from the human mean to the variance among human judges.

LLM	Prompting strategy	Case 1			Case 2		
		MAD	Levene statistic	<i>p</i> -Value	MAD	Levene statistic	<i>p</i> -Value
GPT	Zero-Shot	0.316	16.0	<0.01	0.476	12.0	<0.01
	CoT	0.340	18.5	<0.01	0.497	15.9	<0.01
	Few-Shot	0.266	21.7	<0.01	0.468	20.5	<0.01
Gemini	Zero-Shot	0.430	22.8	<0.01	0.307	45.6	<0.01
	CoT	0.414	27.2	<0.01	0.367	36.2	<0.01
	Few-Shot	0.439	12.7	<0.01	0.427	24.3	<0.01
Claude	Zero-Shot	0.222	32.9	<0.01	0.297	33.9	<0.01
	CoT	0.166	46.9	<0.01	0.336	29.3	<0.01
	Few-Shot	0.266	27.3	<0.01	0.769	13.7	<0.01
Human Judges		0.723	–	–	0.942	–	–

Table 5. Case 1—Facts changed.

LLM	Prompting strategy	Original case		Modified case		U statistic	<i>p</i> -Value
		Average	Standard deviation	Average	Standard deviation		
GPT	Zero-Shot	2.3666	0.4138	1.0833	0.2266	893.5	<0.01
	CoT	2	0.2936	1.4167	0.4099	757.5	<0.01
	Few-Shot	2	0	1.7	0.2449	720	<0.01
Gemini	Zero-Shot	2.5	0	1.8	0.2449	900	<0.01
	CoT	2.5	0	2.0667	0.4955	645	<0.01
	Few-Shot	2.2833	0.252	1.7333	0.4326	759.5	<0.01
Claude	Zero-Shot	2	0	1.3333	0.2357	900	<0.01
	CoT	2	0	1	0	900	<0.01
	Few-Shot	2.5	0	1.1833	0.2409	900	<0.01

Means, standard deviations (SD), and Mann–Whitney comparisons between the original and modified vignettes (30 runs per condition).

differences in distributions between the original and modified cases. Importantly, the direction of change is consistent across models: average sentences decrease in the modified case (for example, GPT Zero-Shot moves from 2.36 years in the original case to 1.08 years in the modified case).

These results demonstrate that LLMs are not simply repeating static sentencing patterns. Instead, they respond systematically to changes in the factual record. This fact-dependence is desirable from a legal standpoint, but it also highlights sensitivity to input framing, which should be carefully managed to avoid biased or manipulated outcomes.

Discussion

This study evaluated whether large language models (LLMs) can function as effective judicial decision-makers by comparing their sentencing outputs to those of 123 retired judges in two fictional criminal cases. The comparison focused on two core dimensions of sentencing quality: consistency and accuracy. Consistency was defined as the internal variance in sentencing decisions for the same case; accuracy was assessed by comparing how closely each model's output, as well as each judge's sentence, aligned with the average sentence imposed by human judges—treated here as a conservative proxy for the correct sentencing decision.

The results provide strong empirical support for the claim that LLMs exhibit significantly greater internal consistency than human judges. Across nearly all models and prompting strategies, the standard deviation of LLM-generated sentences was markedly lower than that of the judges. This result reinforces longstanding concerns about sentencing disparity in human decision-making, as first emphasized by Frankel (Frankel, 1973) and subsequently validated in empirical studies (Austin and Williams, 1977; Clancy et al., 1981; Partridge & Eldridge, 1974). Our findings confirm that even when presented with identical case facts, human judges vary substantially in their sentencing decisions, while LLMs remain highly stable.

More strikingly, the study shows that this increased consistency does not come at the expense of accuracy. In the first case, the average sentence produced by most LLM configurations was statistically indistinguishable from the human average. In the second case, some LLMs produced slightly harsher sentences, yet their decisions were still closer to the human average than most individual judges. This suggests that, under the conservative assumption that the human average is a valid benchmark, LLMs not only match but may outperform human judges in terms of proximity to the consensus.

Furthermore, when we relax the assumption that the average human sentence reflects the correct outcome, the advantage of LLMs becomes even clearer. Because human judges are widely dispersed around the mean, their deviations from the actual correct sentence (if it differs from the mean) are, by definition, larger than their deviations from the mean. In contrast, the narrow distribution of LLM outputs ensures that they remain closer to any fixed reference point, including the correct one. Thus, LLMs demonstrate both superior consistency and greater accuracy relative to individual human judges, regardless of what the “correct” sentence is.

We also examined variation across different LLMs and prompting strategies. Although no single model or strategy proved universally superior, Gemini in the few-shot setting consistently aligned closely with human averages in both cases. This highlights the potential of example-based prompting to improve LLM calibration in legal contexts. Still, given the limited number of case vignettes, we caution against drawing definitive conclusions about specific model-strategy pairings.

To assess the effect of the optimal temperature parameter, set to zero in this research, we note that this configuration represents the most stable operational mode of current LLMs. A temperature of zero does not make the model fully deterministic. Still, it substantially reduces stochastic variability in generated responses, ensuring that differences across models and prompting strategies reflect systematic rather than random variation. This approach is consistent with recent empirical studies across domains that evaluate model reliability under different temperature settings. For instance, Renze (2024) showed that reducing the sampling temperature improves problem-solving reproducibility in LLMs without impairing reasoning accuracy in natural language and logic tasks. Similarly, Patel et al. (2024) examined temperature effects in clinical text interpretation and found that model accuracy and diagnostic reliability remain stable across temperature ranges. Li et al. (2025) analyzed multiple model families and demonstrated that large-scale models are relatively resilient to temperature variation.

In this context, our use of a zero-temperature configuration was intended to isolate intrinsic reasoning stability rather than to suppress all variability. Nevertheless, because most users interact with LLMs with a default temperature of one, we replicated all experiments under these conditions. As shown in Table 4, even at default temperatures, all models outperformed human judges in sentencing accuracy. As detailed in Tables C1 and C2 of Appendix C, most prompting strategies yielded similar performance in terms of consistency across temperature settings. These findings indicate that the models' consistency and accuracy are not artifacts of hyperparameter selection but rather reflect robust, generalizable patterns of LLM behavior across sampling regimes.

While these technical findings highlight the stability of LLMs, it is important to contextualize this performance within the broader stages of the sentencing process. A significant body of sentencing literature suggests that the "In/Out" decision—whether to incarcerate a defendant or impose a community sanction—is a primary site for the manifestation of extra-legal disparities, such as those based on race, gender, and socioeconomic status (Spohn, 2000). Critics of current sentencing practices rightly point out that these binary thresholds often serve as the first point of entry for systemic bias.

However, our study intentionally focused on the length of custodial terms to isolate a different, yet equally pervasive, phenomenon: "noise" or unexplained sentence inconsistency. While systemic bias reflects predictable (though unjust) differences between groups, noise reflects the inconsistencies arising from differences in judicial philosophy or even situational factors (Kahneman et al., 2021). As our results demonstrate, even when the "In/Out" decision is held constant, human judges exhibit

substantial variability in the quantum of punishment, indicating that noise remains a substantial challenge to sentencing uniformity.

Adapting LLMs to the “In/Out” threshold presents unique challenges, as these decisions often hinge on qualitative assessments of rehabilitative potential and public risk. While the ability of LLMs to process narrative data offers a potential advantage over rigid actuarial tools, future research must examine whether such deployment scales systemic biases up or down. Unlike humans, AI systems can be systematically audited and reprogrammed to ignore specific variables, yet they also risk institutionalizing historical prejudices if not governed by robust legal standards.

The tendency of LLMs to cluster around six-month increments might initially appear as a form of “crude reductionism” that overlooks individual nuances. However, our data suggest that this “reductionism” reflects prevailing judicial norms; roughly 85% of the retired judges in our sample also imposed sentences in half-year increments. Hence, the LLMs’ choice of six-month increments represents a replication of common judicial behavior rather than a mechanical limitation.

LLMs technology is evolving rapidly, with each new generation demonstrating notable improvements in contextual reasoning, factual grounding, and adaptability to textual inputs. The models examined in this study were publicly available at the time of data collection and thus reflect a specific technological moment. More recent LLMs may display enhanced capabilities, including greater sensitivity to complex legal reasoning, moral framing, and subtle contextual cues embedded within case narratives. Accordingly, our findings should be viewed as a descriptive benchmark of LLM performance under the conditions and model versions accessible during the study period. Recognizing this dynamic, we aim for this study to serve as a baseline for understanding the trajectory of LLM capabilities in judicial decision-making contexts and to motivate continuous, comparative evaluation as models evolve.

Future research should extend these findings to a broader array of cases and sentencing contexts, explore the robustness of LLMs to subtle factual variations, and assess whether LLMs can generalize across offense types, legal systems, and normative frameworks. More broadly, the integration of AI into sentencing practices raises pressing normative and institutional questions: Who defines the training data, reference norms, and legal standards that shape AI outputs? And how should courts balance the benefits of consistency and efficiency with concerns about transparency, contestability, and human dignity?

Overall, this study contributes to the emerging literature on AI in legal decision-making by comparing LLMs and human judges’ sentencing in the same cases. The results indicate that LLMs can replicate—and in some respects exceed—human performance on key metrics of sentencing quality. These findings warrant careful consideration in ongoing debates about the role of AI as a decision-support tool in the justice system.

Future Implications and Ethical Considerations

Our findings—that LLMs generate sentencing decisions with greater consistency and, under conservative assumptions, greater accuracy than individual judges—raise

important normative and institutional questions about the role of AI in criminal justice. If LLMs can reduce unwarranted disparities without compromising legal standards, they may help address persistent problems of fairness and inefficiency in judicial systems.

However, integrating AI into sentencing also poses significant challenges. One concern is the transfer of normative authority: sentencing decisions involve moral judgment, accountability, and the imposition of punishment, all of which traditionally rest on human discretion. Delegating this authority to machines risks eroding public trust and blurring lines of responsibility (Thomsen, 2022).

A second challenge involves bias and fairness. While LLMs may avoid some of the cognitive biases of human judges, they also learn from historical data, which may reflect structural inequalities (Bagaric et al., 2022; Chouldechova, 2017). Without careful oversight, these biases can be encoded, scaled, and rendered less transparent.

Third, the issues of transparency and explainability remain a challenge. In our study, neither human judges nor LLMs were asked to explain their sentencing decisions. However, LLMs can generate step-by-step rationales when prompted to do so (e.g. *via* chain-of-thought prompting). While this offers an advantage over many traditional algorithms, such explanations are often generated after the fact and may not reflect the model's true internal reasoning (Chiao, 2022; Ryberg & Petersen, 2022). This raises concerns about whether such rationales satisfy legal standards of transparency, accountability, and contestability.

A widely proposed compromise is the use of AI as a decision-support tool, or "co-pilot," rather than a replacement for human judges (Ryberg, 2025; Schwarze & Roberts, 2022). Yet this model presents its own risks. Studies in other domains show that human-AI teams often underperform relative to either component alone, due to overreliance or undertrust (Agarwal et al., 2023; Goh et al., 2024). Moreover, decision support does not necessarily reduce case backlogs, costs, or error rates (Ho et al., 2023; Stevenson & Doleac, 2024).

These concerns underscore the need for robust institutional safeguards, legal standards for AI transparency, and clear rules of accountability. Future research should explore not only whether LLMs outperform human judges but also how and under what conditions they can be responsibly integrated into judicial workflows.

Limitations and Future Directions

This study has several limitations that merit consideration. First, we examined only two fictional case vignettes, which limits the generalizability of our findings and precludes a systematic exploration of how model type and prompting strategy perform across diverse judicial scenarios. However, prior vignette-based research consistently shows that human sentencing decisions are highly variable across a wide range of cases, and there is little reason to expect LLMs to be less consistent if scenario details were altered. Additionally, the consistency and accuracy demonstrated by the LLMs in our study are unlikely to be specific to these particular cases, as the findings were consistent across different models and prompting strategies.

Second, while the use of retired judges provides a rare and valuable dataset, it may not fully reflect the behavior of currently serving judges who operate under different institutional pressures and caseload demands. However, previous studies involving sitting judges have revealed similarly large disparities in sentencing decisions.

Third, the sentencing decisions were made in a controlled, vignette-based setting rather than in actual courtrooms, which typically involve more complex and dynamic information flows. Although this might appear to be a limitation, it may actually understate the inconsistency of human decisions. In real-world judicial environments, additional sources of “noise”—such as fluctuations in mood, time of day, interpersonal interactions, or the appearance and demeanor of defendants—can exacerbate inconsistency (Kahneman et al., 2021). Thus, LLMs may offer an even greater consistency advantage in real-world settings than observed here. Future studies should broaden the dataset to include a wider variety of case types, decision-makers, and judicial environments, and explore ways to establish more reliable and normatively grounded benchmarks for sentencing accuracy. Such studies can also examine cases where the outcomes include non-custodial sanctions or combinations of different types of penalties, in addition to custodial sentences.

An additional limitation is the study’s focus on internal consistency—the extent to which a decision-maker, whether human or machine, produces stable sentencing outcomes when presented with identical case descriptions. We did not examine robustness, defined as the stability of sentencing decisions when minor variations are introduced in the wording, framing, or sequencing of case facts. Prior work shows that LLMs can be sensitive to such variations, occasionally yielding divergent outputs even when the underlying facts remain constant (Zhuo et al., 2024). However, human judges are also known to be affected by such framing effects and cognitive heuristics (Engel, 2022; Peer & Gamliel, 2013). Future research should directly compare the robustness of LLMs and human judges by systematically manipulating case presentation and evaluating the resulting variability in sentencing. A full assessment of both internal consistency and robustness is crucial to evaluating the reliability and suitability of LLMs for judicial decision-making.

Conclusion

This study offers empirical evidence that large language models (LLMs) can generate sentencing decisions that are not only more consistent but, in many configurations, also more closely aligned with prevailing sentencing practices, as approximated by the average sentence imposed by human judges, than those of individual judges. Analyzing responses from 123 retired judges to two fictional case vignettes, we found substantial variability in sentencing—standard deviations of 0.87 and 0.92—despite identical factual inputs. In contrast, LLMs demonstrated markedly lower variance across nearly all models and prompting strategies. This consistency is not merely a technical feature but a normative asset: reducing unwarranted sentencing disparities is a core aim of many legal reforms and an important dimension of justice.

Although LLMs were not always perfectly aligned with the average human sentence, their outputs were, on average, closer to the judges’ mean than were the decisions of the judges. Nevertheless, we do not argue for the immediate adoption of LLMs in

judicial practice. Current models remain limited in transparency, contestability, and normative grounding. Moreover, their deployment in legal contexts raises profound ethical, institutional, and democratic questions. Who defines the standards by which such models operate? How can their use be governed to safeguard fundamental rights and public trust? These are not purely technical challenges but societal ones.

Future work must grapple with both the technological and normative dimensions of AI in law. As LLMs continue to evolve, so too must our understanding of how they might be integrated into legal systems—not as replacements for human judgment, but as tools whose design and governance reflect the values of justice they are meant to serve.

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CRedit: **Oren Gazal Ayal**: Conceptualization, Data curation, Methodology, Project administration, Writing – original draft; **Zohar Elyoseph**: Conceptualization, Investigation, Writing – review & editing; **Adir Solomon**: Data curation, Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing.

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Appendix A: The questionnaire

For research on consistency in sentencing, conducted under the guidance of Prof. Oren Gazal-Ayal, we need your assistance in answering the following questionnaire, which is distributed exclusively to retired judges. To clarify, the research is anonymous; participants' names will not be disclosed even to us and therefore will not be published, and the questionnaire will only be used for statistical analysis of the data and findings. Thank you very much for your willingness to participate.

Two scenarios are presented, describing sentencing arguments in different cases. Please indicate what sentence you would impose on the defendants in these cases (we understand that the data is presented in summary—please do your best to determine the sentence you would impose based on the existing data. There is, of course, no correct answer. Please rely only on the sentencing arguments detailed in the questionnaire; there is no need to search for case law or other materials).

Case 1

Facts of the Indictment and Conviction. Assaf, a pilot cadet, was spending his leave with his girlfriend Rotem at the "Library" pub in Haifa. Dudi, a Golani combat soldier who was also at the pub, saw Rotem and whistled a "sexist" whistle toward her. Assaf, who was somewhat drunk, lunged at Dudi and struck him in the face and teeth with his fists, and a violent brawl broke out between the two. After a few minutes, Dudi fell to the ground, and Assaf broke the beer bottle he was holding and stabbed Dudi in the chest and abdomen with the shard until Dudi stopped moving. As a result of these actions, Dudi suffered severe cuts to his abdomen and chest, requiring stitches under general anesthesia, as well as blunt trauma and sensitivity in the area. In addition, Dudi suffered a bleeding cut on his cheek and a fracture in two of his front teeth. Dudi was hospitalized for eight days.

The Plea and Probation Service Report. Assaf was convicted based on his guilty plea of causing grievous harm under aggravating circumstances—an offense according to Section 333 + Section 335(a)(1) of the Penal Law, 1977.

The probation service report prepared for Assaf revealed that this entanglement with the law had very severe consequences on his life, his military service, and his general functioning. On the evening of the incident, Assaf had argued with his girlfriend, and this heated argument, combined with alcohol, caused him to lose his composure and act violently for the first time in his life. The probation officer noted that Assaf expresses remorse for his actions and still hopes that he can return to the pilot cadet course (from which he was dismissed following the filing of the indictment). The report also stated that this is a serious person who was an exemplary cadet and who is suitable for therapeutic treatment, and therefore, the recommendation was to settle for as moderate a prison sentence as possible, allowing him to rehabilitate his life upon completion of his sentence.

Prosecution's Sentencing Arguments. The prosecution argued that given the severity of the actions, the defendant should be sentenced to four years of actual imprisonment in addition to a suspended sentence. In its sentencing arguments, the prosecution presented the following case law:

1. Criminal Case (Haifa District) 3195/04 State of Israel v. Manshirov—Following a guilty plea the defendant was convicted of causing grievous harm under aggravating circumstances. Following an argument over a trivial matter, the defendant stabbed the complainant twice in the chest with a knife. The complainant suffered a severe injury to the diaphragm and spleen, was hospitalized, and underwent surgery. Although this was the defendant's first conviction, and the probation service report was positive, the court sentenced him to four and a half years of actual imprisonment, eighteen months of suspended imprisonment, and compensation of 15,000 NIS, due to the severity of the act and the fact that only one

millimeter in the depth of the knife's penetration separated the injury from causing the complainant's death.

2. *Crim. App. 6260/05 Chadra v. State of Israel*—The appellant was convicted based on a guilty plea of causing injury and harm under aggravating circumstances, after, in a violent brawl that erupted over a family dispute, the appellant drew a sharp object and stabbed the complainant in the abdomen. The complainant was rushed to the hospital and had to undergo surgery. The appellant has no criminal record, but the probation service found that the appellant has a childish personality, grew up in conditions of emotional neglect, and is not suitable for rehabilitation. Therefore, the probation service did not make a therapeutic recommendation. The Supreme Court chose not to intervene in the sentence imposed by the District Court—three years of actual imprisonment and an additional year of suspended imprisonment. The decision stated that a sentence of three years of actual imprisonment for a stabbing that caused an injury requiring hospitalization and surgery is not a severe sentence at all, and that due to the multitude of cases of using cold weapons to resolve trivial disputes, there is no choice but to take a strict approach, even though this concerns a young person with no criminal record.

Defense's Sentencing Arguments. Assaf's defense attorney argued that he should not be dealt with harshly since this is his first entanglement with the law. He noted that the defendant is young and has his whole life ahead of him, and that the conviction will anyway lead to his dismissal from the pilot cadet course and a severe stain that will accompany him throughout his life. The defense attorney also noted that the defendant cooperated with the investigation, regretted and repented his actions, and is even willing to compensate the complainant. He asked to settle for one year of imprisonment and presented the following case law to support his claims:

1. *Crim. App. 45636-06-13 State of Israel v. Tal*—The defendant was convicted based on his guilty plea of assaulting another by hitting his head with a bottle twice until the bottle broke on the complainant's head, and then smashing the broken bottle into the complainant's face, causing a deep cut about 10cm long, as well as two cuts on his arm, which required medical treatment. Given the defendant's young age and clean record, and based on the positive probation service report, but also taking into account the severity of the act, the court sentenced the defendant to one year of actual imprisonment.
2. *Criminal Case (Haifa District) 51486-05-14 State of Israel v. Murad*—The defendant was convicted based on his confession of offenses of causing grievous harm under aggravating circumstances. During a brawl, the defendant attacked the complainant with a sharp object and caused him cuts in the left chest, neck, head, and cheek, hematomas and signs of injury to the chest, as well as a fracture in one of the fingers. It was determined that the injury was moderate, the certain reciprocity in initiating the incident constituted a mitigating factor, and in addition, in recent years, the defendant had led a normative life and had not been involved in criminal activities. The court also decided to take into account the guilty plea and the saving of judicial time, and therefore sentenced the defendant to eighteen months of actual imprisonment, twelve months of suspended imprisonment, and compensation to the complainant in the amount of 12,000 NIS.

Under these circumstances, what sentence would you impose on the defendant?

Case 1

Facts of the Indictment and Conviction. As part of a plea bargain, an amended indictment was presented to the court, describing the following agreed facts:

Eyal, an accountant, and Dorit, an investment advisor at a bank, both single and twenty-seven years old, have been friends since a young age. The two used to talk on the phone once a week

and meet once every few months. During one of these meetings, at Eyal's home, Eyal momentarily caressed Dorit, then approached her, and they kissed. The contact between them continued consensually until Eyal began to unbutton Dorit's shirt. At this point, Dorit stopped Eyal, told him it was a mistake, and that she wanted to return home. Eyal continued to undress her while telling her that he loved her and tried to persuade her to cooperate, but Dorit, embarrassed by the situation, repeatedly asked him to stop, while gently trying to push his hands away from her and prevent him from continuing his actions. Eyal was not deterred, continued to undress her, until Dorit ceased her physical resistance and just cried and asked him to stop. Eventually, while she was crying, and as he continued to tell her words of love, Eyal had sexual intercourse with Dorit.

In the original indictment, Eyal was also charged with an indecent act (due to his actions preceding the rape), but as part of the agreement, this section was removed from the indictment, and Eyal was convicted of rape based on guilty plea. The plea bargain did not include an agreement regarding the sentence.

Position of the Victim and Probation Service. Dorit did not suffer physical injury as a result of the act. She requested that a victim impact statement not be prepared for her and that she not be involved in the criminal proceedings, as she wishes to put the matter behind her and return to her life.

The probation service report prepared for Eyal indicates that this entanglement with the law was traumatic for him. Upon his arrest, which lasted three months, Eyal lost his job. The probation officer noted that Eyal expresses remorse for his actions, although sometimes it emerges from the conversation that he feels that Dorit is also to blame for hinting that she was interested in him, for deciding to come to his home, and for kissing him. The probation service recommended moderate sentencing that would allow Eyal to rehabilitate his life.

Prosecution's Sentencing Arguments. In the sentencing arguments, the prosecution presented the following case law and requested a sentence of five years of imprisonment:

1. Crim. App. 54/11 Epenger v. State of Israel—The defendant and the complainant, who had a prior brief acquaintance, met at a club, drank alcoholic beverages together, and kissed. When the defendant tried to touch the complainant's body, she resisted and asked him to stop, but despite her resistance, he tried again and again. When the complainant went to the women's restroom, the defendant accompanied her, and when she tried to close the cubicle door, he pushed the door, entered the cubicle after her without her consent, locked the door from the inside, and forcibly turned the complainant around, with her face to the wall and her back to him. In this position, the defendant inserted his hand under the complainant's underwear and inserted his fingers into her genitals and anus. Subsequently, despite her resistance, he pressed her against the wall, slightly lowered her underwear, and slightly inserted his penis into her anus for a few seconds. The Supreme Court rejected the defendant's appeal. It stated that the lower court had considered the defendant's personal circumstances (no criminal record) when sentencing, and that the sentence imposed even tended toward leniency, so there was no place to intervene. The defendant was aware of the complainant's refusal but did not let it stand in his way, and he also did not take responsibility for his actions. Therefore, the Supreme Court confirmed the sentence imposed on him in the District Court—three years of actual imprisonment, twelve months of suspended imprisonment, and compensation in the amount of 30,000 NIS.
2. Criminal Case (Tel Aviv District) 1023/03 Vaknin v. State of Israel—The defendant was the complainant's regular taxi driver, and on the day of the incident, he entered her home after driving her, where he tried to hug her, but the complainant tried to evade him. The defendant grabbed her hands forcefully and hugged her, and subsequently dragged her up the stairs to the bedroom, pushed her onto the bed and lay on top of her, undressed her of her pants and underwear, undressed himself, and inserted his penis into her genitals, ignoring her resistance and screams, until he reached sexual satisfaction. The defendant's version, according to which the relations took place with the complainant's consent, was rejected, and after hearing evidence, he was sentenced to five years of actual imprisonment, suspended imprisonment for two years, and compensation to the complainant in the amount of 10,000 NIS.

Defense's Sentencing Arguments. Eyal's defense attorney argued that Eyal should not be dealt with harshly, as this is his first entanglement with the law. He noted that the defendant is young and has his whole life ahead of him, and actual imprisonment would lead to his dismissal and even prevent his future employment. The defendant cooperated with the investigation, admitted his actions, regretted and repented, and is even willing to compensate the complainant. Therefore, the defense attorney asked to settle for a sentence of fifteen months of actual imprisonment.

The defense attorney argued that the examples brought by the prosecution are too severe and presented the following case law:

1. Crim. App. 307/11 Sharon v. State of Israel—In this case, the defendant, a professional masseur, was convicted of raping a patient, when during the treatment he moved her underwear and inserted his finger into her genitals without her consent. The complainant immediately pushed his hand away, got dressed, and left the treatment room. The court noted the severity of the offense, pointing to the breach of trust given to the defendant as a certified masseur on the one hand, and to the mitigating factors on the other hand: the defendant has no criminal record, he volunteered for the army despite receiving an exemption, he experienced trauma during the arrest and was fired from his job as a result. The opinion claimed a low level of sexual danger. The court sentenced him to fifteen months of actual imprisonment, suspended imprisonment, and compensation to the complainant.
2. Crim. App. 4876/15 John Doe v. State of Israel—The complainant and the defendant were a couple for a long period, but before the case in question, a dispute arose between them. On the day of the incident, the two met by chance at a nightclub. The defendant forcibly led the complainant to his home, laid her on the bed with one hand over her mouth, and with his other hand lowered his pants and underwear. He lifted her shirt to her chest and inserted his hand from above. The complainant resisted, went wild, screamed, and hit him. After he failed to carry out his plan, the defendant held her with his hand while she was lying on the bed and with his other hand masturbated until he reached satisfaction and ejaculated on her stomach. The appellant was convicted of attempted rape under Section 345(a)(1) of the Penal Law. In the probation service report, it was written that the appellant takes responsibility for committing the offense and recognizes the element of coercion in his actions, and that he expresses empathy toward the complainant and recognizes his harm to her, which serves to reduce the risk of committing offenses in the future. He is also being treated in a psycho-educational group for sex offenders. The appellant married and started working regularly, and his criminal involvement has moderated. Because of all these factors, the court sentenced him to seventeen months of actual imprisonment.

Under these circumstances, what sentence would you impose on the defendant?

Personal Details

Thank you very much for your response. In closing, we would appreciate some personal details:

- Gender: Male/Female
- Age:
- Last court in which you served: Magistrate's/District/Supreme/Other
- During your tenure, you sentenced defendants: Never/Few times/Many times
- Before the judicial role, you served in the following positions (you can mark more than one answer): Defense attorney/Prosecutor or criminal attorney/Civil attorney/Other

Thank you very much!

Appendix B: Reference cases

This appendix presents the complete set of cases used as illustrative examples in the few-shot prompting technique.

Violent Offense

Case 1. The defendant was convicted of aggravated assault under Sections 333 and 335(a)(1), and threats under Section 192 of the Penal Code. According to the amended indictment, following an argument at a bar in Beit Shemesh, the defendant cursed the complainant, threw objects at him, and then stabbed him in the face and neck with metal tongs. The complainant required surgery and was hospitalized for seven days, resulting in permanent vision loss in his right eye.

The conviction followed the defendant's admission of guilt as part of a plea agreement that did not include sentencing terms.

Aggravating Circumstances:

- Use of a cold weapon in a public place
- Severe injuries, including permanent vision loss
- Defendant initiated and pursued the confrontation
- Significant functional and psychological harm to the complainant
- Minimization of the offense's severity to probation services

Mitigating Circumstances:

- Admission of guilt and genuine remorse
- No prior criminal record (except for a pending case)
- Full participation in therapy
- Ceased alcohol consumption and committed to treatment
- Difficult family and financial circumstances
- Time served under house arrest and electronic monitoring

Sentence Imposed:

- 9 months of community service
- 12 months suspended sentence (3 years) for violent felonies
- 6 months suspended sentence (3 years) for violent misdemeanors
- Compensation of NIS 30,000 to the complainant
- 12 months probation

The court determined a sentencing range of 20–40 months but deviated downward due to the defendant's significant rehabilitation process.

Case 2. The defendant was convicted of aggravated assault under Sections 333 and 335(a)(1) and possession of a knife under Section 186(a) of the Penal Code. According to the amended indictment, the defendant entered a gas station convenience store carrying a knife. Following an argument and an attempted theft of a tuna can, he stabbed the complainant in the abdomen and fled while making threats. The complainant underwent emergency surgery due to an intestinal injury and was hospitalized for five days.

The conviction was entered during trial based on a plea agreement that acknowledged proximity to the insanity defense.

Aggravating Circumstances:

- Knife attack causing severe injury requiring surgery
- Fleeing the scene while threatening the victim
- Assault on a convenience store worker at night
- Extensive criminal record (18 prior convictions)
- Lack of responsibility or remorse

Mitigating Circumstances:

- Admission during trial, sparing the complainant's testimony
- Mental illness and psychotic state close to legal insanity
- Harsh life circumstances, including homelessness and psychiatric illness
- Particularly difficult detention conditions

Sentence Imposed:

- 28 months of actual imprisonment
- 10 months suspended sentence (3 years) for violent felonies
- 5 months suspended sentence (3 years) for violent misdemeanors or knife possession
- Compensation of NIS 7,500 to the complainant

The court set a sentencing range of 18–48 months, balancing mental health considerations with the severity of the offense and public interest in combating knife violence.

Case 3. The appellant was convicted of aggravated assault under Sections 333 and 335(a)(1) and threats under Section 192 of the Penal Code. According to the amended indictment, the appellant threatened the complainant, returned with a metal rod, and forcefully struck him, fracturing his left shin. The victim report noted the complainant was left emotionally shattered by the incident.

The conviction was based on the appellant's admission.

Aggravating Circumstances:

- Unprovoked attack
- Exceptional degree of violence
- Severe harm to the complainant
- Criminal history, including prior conviction for serious violence (4 years for stabbing)
- Profound negative impact on the complainant's life

Mitigating Circumstances:

- Participation in therapeutic programs in prison
- Positive probation service evaluations regarding rehabilitation potential

Sentence Imposed:

- 4 years of actual imprisonment
- Compensation of NIS 20,000 to the complainant

Appeal Outcome: The Supreme Court rejected the appellant's claim that the sentence was harsh compared to similar cases and upheld the punishment, citing the severity of the violence, the resulting harm, the appellant's criminal history, and the lack of provocation.

Sexual Offense

Case 1. The defendant was convicted by confession as part of a plea agreement for the offenses of rape and aggravated assault of his spouse.

The conviction followed a plea deal that included an agreement on the facts, but not on the sentence.

According to the amended indictment, the defendant and the complainant were married for about 20 years. In 2014, the defendant raped the complainant after she refused to have sex with him. Additionally, between 2014–2018, he physically assaulted her on several occasions, including shoving, grabbing her neck, and pouring water on her face.

Aggravating Circumstances:

- Severe violation of the complainant's dignity and autonomy in her own home
- Prolonged violent behavior over several years
- Abuse of trust and familial relationship

Mitigating Circumstances:

- Confession and expression of remorse
- No prior criminal record
- Full cooperation with an extended therapeutic process
- Significant rehabilitation and recognition of the gravity of his actions
- Complainant's support for his rehabilitation

A probation report and risk assessment indicated substantial progress in therapy, genuine remorse, and a reduced risk level (from moderate to low-moderate).

Sentence Imposed:

- 9 months of custodial sentence served as community service
- 24 months suspended sentence for sexual/violent felonies
- 7 months suspended sentence for sexual/violent misdemeanors
- 24 months probation
- Compensation of NIS 40,000 to the complainant

The court deviated from the established sentencing range of 3–6 years' imprisonment due to the successful rehabilitation process and high chances of reintegration.

Case 2. The appellant was convicted in the District Court of rape. The conviction followed a full evidentiary trial.

According to the proven facts, the appellant was a taxi driver who occasionally drove the complainant. On the day of the incident, he drove her home and entered her apartment. After drinking coffee together, he tried to approach her and she resisted. He dragged her into the bedroom by force and raped her despite her opposition. The complainant reported the incident to the police that same day after telling her brother.

Aggravating Circumstances:

- Exploiting a relationship of trust between a taxi driver and passenger
- Serious violation of the complainant's autonomy
- Committing the act in the complainant's home, exploiting a power imbalance

Mitigating Circumstances:

- One-time incident
- Expression of regret
- Rehabilitation efforts from criminal and drug history
- No prior convictions for sexual offenses

Sentence Imposed by District Court:

- 5 years of actual imprisonment
- Suspended sentence
- Compensation of NIS 10,000 to the complainant

Supreme Court Ruling: The appeal against both conviction and sentence was unanimously dismissed. The court found the trial court's findings—based on credibility assessments—reliable and saw no reason to intervene. The sentence was deemed a fair balance between the severity of the act and mitigating factors.

Case 3. The defendant was convicted after a full evidentiary hearing of rape and indecent assault. According to the indictment, the defendant committed the acts against the complainant, whom he met that day at a hotel in Eilat. After the complainant left the group and went to rest in her room, the defendant followed her and began touching her body against her will, ignoring her verbal and physical resistance. He inserted his fingers and penis into her and continued until he ejaculated. The defendant denied the charges and claimed that the sexual intercourse was consensual.

Aggravating Circumstances:

- Severe violation of bodily integrity and autonomy
- Planned behavior: followed her to her room
- Blatant disregard for the complainant's resistance
- Significant psychological harm as reflected in the victim impact report
- Criminal history including violent offenses
- Lack of accountability and remorse

Mitigating Circumstances:

- Difficult family circumstances
- Married and father of an infant
- No use of extreme physical violence

Sentence Imposed:

- 8 years of actual imprisonment
- 12 months of suspended sentence
- Compensation of NIS 70,000 to the complainant

The court set a sentencing range of 6.5 to 10.5 years and emphasized the need for strict punishment in sex offense cases.

Appendix C: Temperature varying results

Tables C1 (Case 1) and **C2** (Case 2) present the average sentence length and standard deviation for each LLM under temperature = 0 and the default temperature, together with the Mann–Whitney results.

From the tables, we observe that even at the default temperature settings, all LLMs produce lower standard deviations than human judges. In Case 1, for example, human judges show a standard deviation of 0.878, compared to LLM standard deviations typically below 0.4. A similar pattern is seen in Case 2, where the human judges' standard deviation is 0.922. This indicates that while stochastic sampling increases variation relative to temperature = 0, LLMs remain far more internally consistent than humans.

In several cases, distributions at temperature = 0 and default temperature are statistically indistinguishable. For instance, GPT with Chain-of-Thought prompting in Case 1 shows no difference across temperature settings. Gemini Zero-Shot and CoT also show stability in Case 1. On the other hand, GPT Few-Shot and Gemini Few-Shot in Case 1 display significant differences, and Gemini Zero-Shot in Case 2 shows extreme sensitivity. These results suggest that certain prompting strategies, especially Chain-of-Thought, can stabilize outputs against temperature-induced variability. The temperature analysis shows that LLMs are consistently less variable than human judges, that some prompting strategies are robust to temperature changes, and that prompt design plays a key role in achieving stability in legal reasoning tasks.

Table C1. Case 1—Temperature changed.

LLM	Prompting strategy	Temperature = 0		Default temperature		U statistic	p-Value
		Average	Standard deviation	Average	Standard deviation		
GPT	Zero-Shot	2.3666	0.4138	2.1889	0.3890	553.5	0.09
	CoT	2	0.2936	1.9944	0.3638	457.5	0.90
	Few-Shot	2	0	2.1833	0.3287	315	<0.01
Gemini	Zero-Shot	2.5	0	2.5667	0.2134	390	0.09
	CoT	2.5	0	2.5500	0.1979	405	0.16
	Few-Shot	2.2833	0.2520	2.5028	0.3981	302.5	0.01
Claude	Zero-Shot	2	0	2.1833	0.2409	285	<0.01
	CoT	2	0	2.0500	0.1500	405	0.08
	Few-Shot	2.5	0	2.2667	0.2808	660	<0.01

Means, standard deviations (SD), and Mann–Whitney comparisons between temperature = 0 and default temperature (30 runs per condition).

Table C2. Case 2—Temperature changed.

LLM	Prompting strategy	Temperature = 0		Default temperature		U statistic	p-Value
		Average	Standard deviation	Average	Standard deviation		
GPT	Zero-Shot	3.0833	0.1895	2.8806	0.5876	612.5	<0.05
	CoT	2.9333	0.2537	2.8444	0.5836	547	0.31
	Few-Shot	2.8166	0.3824	2.6361	0.513	585.5	0.12
Gemini	Zero-Shot	3	0	2.7167	0.3578	765	<0.01
	CoT	3.0666	1.0148	2.6778	0.3966	540	0.39
	Few-Shot	2.5	0	2.6194	0.4593	420	0.33
Claude	Zero-Shot	3	0	3.0667	0.2809	450	0.42
	CoT	2.7833	0.2841	3.0667	0.3091	281	<0.01
	Few-Shot	3.7666	0.2537	3.5	0.4655	642	0.01

Means, standard deviations (SD), and Mann–Whitney comparisons between temperature = 0 and default temperature (30 runs per condition).